

Audio course on steroids during pregnancy. Episode two. Placental transfer and eleven beta H S D two.

Companion reading handout for simultaneous listening.

In the first episode, we built the target-first model. Steroids in pregnancy only make sense after you ask who the target is: fetus, mother, or adrenal axis.

Now we need the pharmacology that makes that model work.

The placenta is not a passive filter. It is a living endocrine and metabolic organ. It decides, partly through blood flow and diffusion, but also through enzyme activity, how much active steroid reaches the fetus.

For corticosteroids, the key enzyme is eleven beta hydroxysteroid dehydrogenase type two. In spoken clinical shorthand, eleven beta H S D two.

This enzyme converts active cortisol-like steroids into less active forms. It is one of the ways the placenta protects fetal tissues from excessive maternal glucocorticoid exposure.

That protection matters because glucocorticoids are not only anti-inflammatory drugs. They are developmental signals. They influence fetal maturation, fetal growth, the fetal H P A axis, brain development, metabolism, vascular biology, and gene expression. The fetus needs glucocorticoid signals, but not unlimited maternal exposure.

So when you choose a steroid in pregnancy, you are partly choosing how much you want to bypass or respect that placental protection.

Clinically, this creates two useful families.

First, the lower fetal-transfer maternal steroids: prednisone, prednisolone, hydrocortisone, and methylprednisolone.

Second, the high fetal-transfer fluorinated steroids: betamethasone and dexamethasone.

The distinction is not absolute. No drug lives in a magical separate compartment. But it is the practical rule that makes sense of most decisions.

Prednisone and prednisolone are useful for maternal disease because the placenta inactivates a large proportion before it reaches the fetus. Hydrocortisone is also substantially metabolized by placental eleven beta HSD two, which is why it makes physiologic sense for maternal replacement. Methylprednisolone behaves similarly enough that it is used for maternal inflammatory disease, including situations where intravenous treatment is needed.

Betamethasone and dexamethasone are different. They cross the placenta efficiently. For fetal lung maturation, that is exactly why we choose them. If the fetal lung is the target, a drug that does not reach the fetus well is the wrong tool.

Pause here, because this explains a common bedside confusion.

A patient is taking chronic prednisolone for autoimmune disease. Now she develops preterm contractions at thirty weeks and looks likely to deliver. Does the chronic prednisolone count as an antenatal steroid course?

Usually, no.

The reason prednisolone is attractive for maternal disease is the same reason it is not a reliable fetal lung maturation course. The placenta reduces fetal exposure. If the clinical indication is fetal lung maturation, the usual fetal therapy is betamethasone or dexamethasone.

Now take the opposite scenario.

A pregnant patient has a lupus flare or inflammatory bowel disease flare. Should we choose dexamethasone because it is potent and crosses the placenta?

Usually, no.

If the disease target is maternal, fetal transfer is not an advantage. Prednisone, prednisolone, or methylprednisolone usually match the target better. The drug should treat the mother while avoiding unnecessary fetal exposure.

That is the pharmacologic heart of the course.

High fetal transfer is not good or bad. It is correct or incorrect depending on the target.

This also helps us understand specialized exceptions.

Dexamethasone may be considered in fetal therapy contexts because it reaches the fetus. Examples include controversial treatment of selected fetal cardiac conduction disease in anti-Ro or anti-La pregnancies, or historically prenatal treatment

discussions around congenital adrenal hyperplasia. Those are not routine maternal steroid decisions. They are intentional fetal exposure decisions, and they require specialist counseling because the fetal benefit must justify the fetal exposure.

The same logic applies to betamethasone. Betamethasone is not a general pregnancy steroid. It is a fetal maturation drug when preterm birth is likely in the appropriate window.

Now let us connect this to dose and duration.

A short course of betamethasone for fetal maturation is a high fetal-transfer exposure, but it is brief and targeted. The benefit in the right early preterm setting is strong.

Chronic systemic steroids for maternal disease are a different exposure pattern. Even if fetal transfer is lower, prolonged maternal dosing can still matter. It can worsen maternal glucose, blood pressure, infection risk, wound healing, mood, bone health, and adrenal suppression. It can also be confounded with the disease itself, because active lupus, severe asthma, active IBD, or severe dermatologic disease may carry fetal risk independent of the drug.

So the correct question is never just, did the fetus see steroid?

The better questions are, which steroid, how much transfer, what timing, what duration, what maternal disease, and what untreated risk?

This is why guidelines can sound different depending on the setting. ACOG guidance on antenatal corticosteroids focuses on deliberately exposing the fetus for fetal maturation. BSR guidance on rheumatologic disease focuses on prednisolone as a compatible and preferred maternal corticosteroid. UKTIS safety guidance focuses on systemic exposure and the difficulty of separating medication risk from underlying disease. These are not contradictory. They are answering different target questions.

Now think like a resident at the bedside.

If a patient with asthma is hypoxic, the fetal problem is maternal oxygenation. A short prednisone course may be safer than undertreated asthma.

If a patient with preterm birth risk needs fetal lung maturation, prednisone is not the right lung-maturation drug.

If a patient has adrenal insufficiency in labor, hydrocortisone is not being given to mature fetal lungs and not being given to suppress a flare. It is replacing a stress response.

If a patient has possible fetal heart block, dexamethasone is not being used because the mother needs a stronger anti-inflammatory drug. It is being considered because fetal transfer is the point, and the evidence and risks must be handled carefully.

That is why placental transfer is not a pharmacology footnote. It is the clinical logic of steroid choice.

The resident reflex is this: ask whether placental crossing is the goal, the unwanted exposure, or irrelevant to replacement physiology.

If placental crossing is the goal, betamethasone or dexamethasone may be right.

If placental crossing is not the goal, prefer a lower fetal-transfer steroid when systemic therapy is needed.

If the issue is adrenal replacement, think hydrocortisone physiology and stress dosing.

The placenta does not make steroid use simple. But it makes steroid use intelligible.